Machine Learning Assignment :- 9

Name: Ghrushnesh Sudhakar Patil

Class: CSAI-D

Batch : 3

Roll No: 66

PRN: 12411658

Q. Write a program to do following:

This dataset gives the data of Income and money spent by the customers visiting a shopping mall. The data set contains Customer ID, Gender, Age, Annual Income, Spending Score. Therefore, as a mall owner you need to find the group of people who are the profitable customers for the mall owner. Apply at least two clustering algorithms (based on Spending Score) to find the group of customers.

DATASET LINK :- https://www.kaggle.com/shwetabh123/mall-customers

A). Apply Data pre-processing

B). Perform data-preparation (Train-Test Split)

C). Apply Machine Learning Algorithm

D). Evaluate Model.

```
from google.colab import files
uploaded = files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Mall_Customers.csv to Mall_Customers.csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
```

```
data = pd.read_csv("/content/Mall_Customers.csv")

data.head()
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

A). Applying Data Pre-Processing

```
encoder = LabelEncoder()
data['Genre'] = encoder.fit_transform(data['Genre'])
```

```
X = data[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

B). Perform data-preparation (Train-Test Split)

```
X_train, X_test = train_test_split(X_scaled, test_size=0.6, random_state=42)

print("Training data size:", X_train.shape)
print("Testing data size:", X_test.shape)
```

```
Training data size: (80, 2)
Testing data size: (120, 2)
```

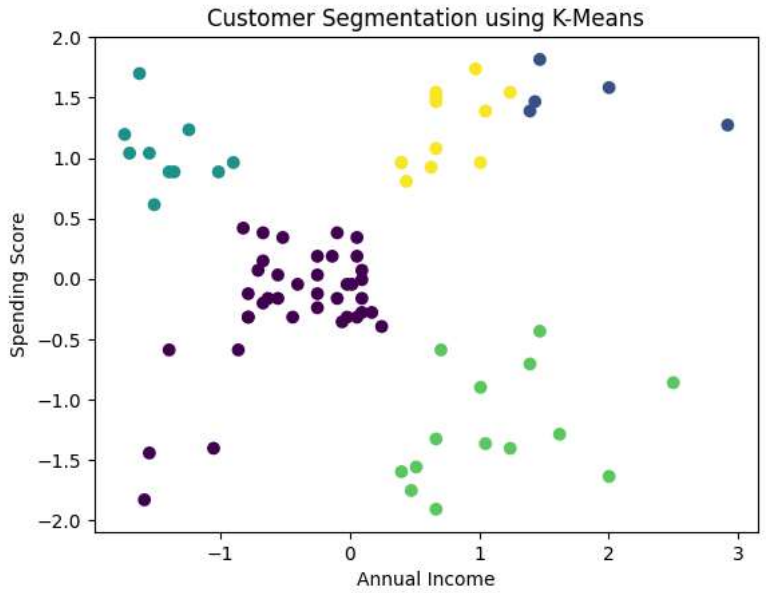
C). Applying Machine Learning Algorithm

K-Means Clustering

```
kmeans = KMeans(n_clusters=5, random_state=42)

kmeans_labels = kmeans.fit_predict(X_train)
```

```
plt.scatter(X_train[:,0], X_train[:,1], c=kmeans_labels)
plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.show()
```

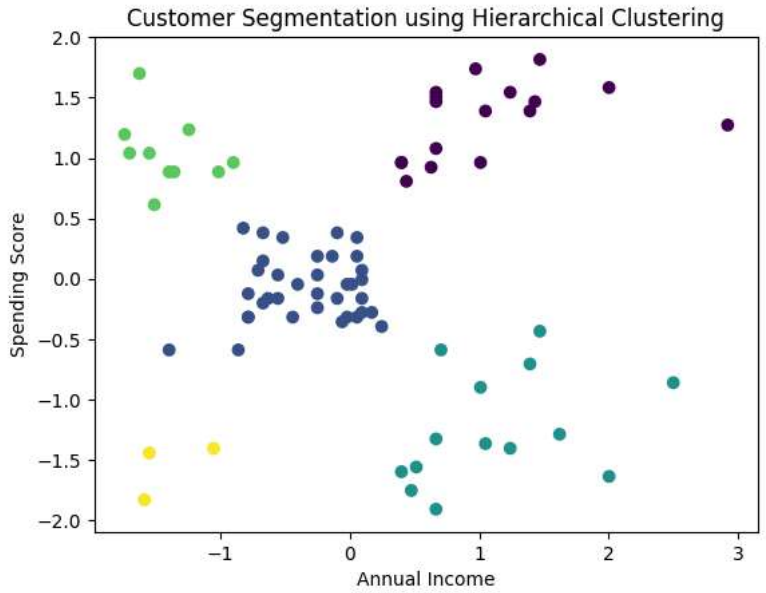


Heirarchial Clustering

```
hc = AgglomerativeClustering(n_clusters=5)

hc_labels = hc.fit_predict(X_train)
```

```
plt.scatter(X_train[:,0], X_train[:,1], c=hc_labels)
plt.title("Customer Segmentation using Hierarchical Clustering")
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.show()
```



D). Evaluate Model

```
kmeans_score = silhouette_score(X_train, kmeans_labels)
hc_score = silhouette_score(X_train, hc_labels)

print("K-Means Silhouette Score:", kmeans_score)
print("Hierarchical Clustering Silhouette Score:", hc_score)
```

```
K-Means Silhouette Score: 0.5020542119325657
Hierarchical Clustering Silhouette Score: 0.5708471227415485
```